

Ahmad Ghalawinji

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PROFILE

MSc graduate in Data Science and AI with a research background in predictive modelling, deep learning, and data engineering. Experienced in building end-to-end pipelines and applying machine learning (gradient-boosted models, VAEs, transformers) to complex, multi-source datasets. Interested in interdisciplinary research bridging climate science, public health, and computational forecasting, with hands-on project experience studying the effects of warming scenarios on youth cardiorespiratory fitness.

EDUCATION

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| MSc — Data Science and Artificial Intelligence (MoSIG M2)
<i>ENSIMAG and Grenoble Alpes University</i>
Relevant coursework: <i>Advanced ML, Natural Language Processing, Big Data Management</i> | Sep 2023 – Sep 2024
<i>Grenoble, France</i> |
| MSc — Informatics (MoSIG M1)
<i>ENSIMAG and Grenoble Alpes University</i>
Relevant coursework: <i>Algorithms, Software Engineering, Operating Systems</i> | Sep 2022 – Sep 2023
<i>Grenoble, France</i> |
| BSc — Computer Engineering
<i>Tishreen University</i>
Relevant coursework: <i>Data Structures, Computer Architecture, Database Systems</i> | Sep 2016 – Sep 2021
<i>Latakia, Syria</i> |

PROFESSIONAL EXPERIENCE

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| Artificial Intelligence Engineering Intern
<i>Amadeus</i> | Feb 2024 – Aug 2024
<i>Nice, France</i> |
| <ul style="list-style-type: none">Improved a production forecasting system by designing a data pipeline in Databricks to integrate external pricing signals into the existing machine learning model.Achieved a 4% improvement in forecast accuracy, resulting in a 0.4% increase in revenue.Technologies: Python, MongoDB, Azure, Spark on Databricks. | |
| Machine Learning Intern
<i>Inria</i> | Feb 2023 – Jul 2023
<i>Grenoble, France</i> |
| <ul style="list-style-type: none">Conducted research into deep neural models such as transformers and variational autoencoders (VAEs) to enhance representation learning.Utilised PyTorch to develop a VAE that reduced data dimensionality, enabling the study of dynamic system behaviour.Technologies: Python, PyTorch, VAEs. | |
| Software Engineering Intern
<i>Laboratoire d'Informatique de Grenoble (LIG)</i> | Feb 2022 – Jul 2022
<i>Grenoble, France</i> |
| <ul style="list-style-type: none">Defined automated planning workflows and implemented the transition from Domain-Specific Language (DSL) to automated planning using a meta-model for Petri nets.Technologies: DSL, Modelling. | |

PROJECTS

Climate Change & Youth Cardiorespiratory Fitness | |

- Investigated how warming scenarios (+1°C to +3°C) affect youth cardiorespiratory fitness across 28 European countries by integrating ERA5 climate reanalysis, World Bank socioeconomic indicators, and WHO physical activity policies.
- Developed a synthetic fitness data generation methodology grounded in exercise physiology literature to demonstrate a complete analytical workflow in the absence of open cross-national youth CRF datasets.
- Trained predictive models (Random Forest, XGBoost) showing a 4.7% baseline CRF decline under +3°C warming; mitigation analysis revealed 55% of climate-driven fitness loss could be recovered through enhanced physical activity policies.
- Built an interactive Streamlit dashboard enabling policymakers to explore climate scenarios and evaluate adaptation strategies in real-time without programming expertise.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, XGBoost, Plotly, Streamlit, Joblib, ERA5 Climate Data, LaTeX.

Port Calls Detection & Vessel Journey Modelling | |

- Built a production-style ETL pipeline to detect vessel port call events from 27,000+ AIS position records across 16 ports, using geodesic geofencing (20 km threshold) and 30-minute deduplication to minimise false positives.
- Orchestrated a two-task Airflow DAG (`PythonOperator`) to extract CSV data into PostgreSQL and detect port entries/exits; inferred next destinations per vessel using a SQL `LEAD()` window function.
- Engineered for reliability: replaced `subprocess` calls with direct Python imports, externalised detection thresholds to environment variables, added structured logging, type hints, and docstrings throughout.
- Improved test coverage with pytest, asserting correct vessel IDs, port names, timestamps, and row values — not just call counts; used `tmp_path` fixtures for realistic CSV integration tests.
- Identified and documented anchoring false positives; proposed mitigation via speed threshold (<0.5 knots) and positional drift analysis to discriminate anchoring from true port docking.
- Delivered an interactive Streamlit dashboard with multi-dimensional filters (vessel, port, event type, date range) and summary metrics; containerised the full stack with Docker Compose.
- **Technologies:** Python, PostgreSQL, Apache Airflow, Streamlit, Docker, Pandas, GeoPy.

SKILLS

Programming: Python (Advanced), SQL, C++, Java

Machine Learning: Supervised & Unsupervised Learning, VAEs, Transformers, Feature Engineering, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch

Data Engineering: Apache Spark, Apache Airflow, Azure Databricks, AWS (S3, EC2), ETL Pipelines, PostgreSQL, PostGIS, MongoDB, ERA5 Climate Data

Visualisation: Streamlit, Plotly

Tools & Deployment: Git, Linux, Docker, LaTeX

LANGUAGES

Arabic: Native

English: Advanced (C2)

French: Intermediate (B2)